LAB EXPERIMENT
DELETION AND RECOVERING OF FILES USING FTK IMAGER

AIM- To demonstrate the digital forensic process of identifying, analyzing, and recovering deleted data from a logical volume using FTK Imager.

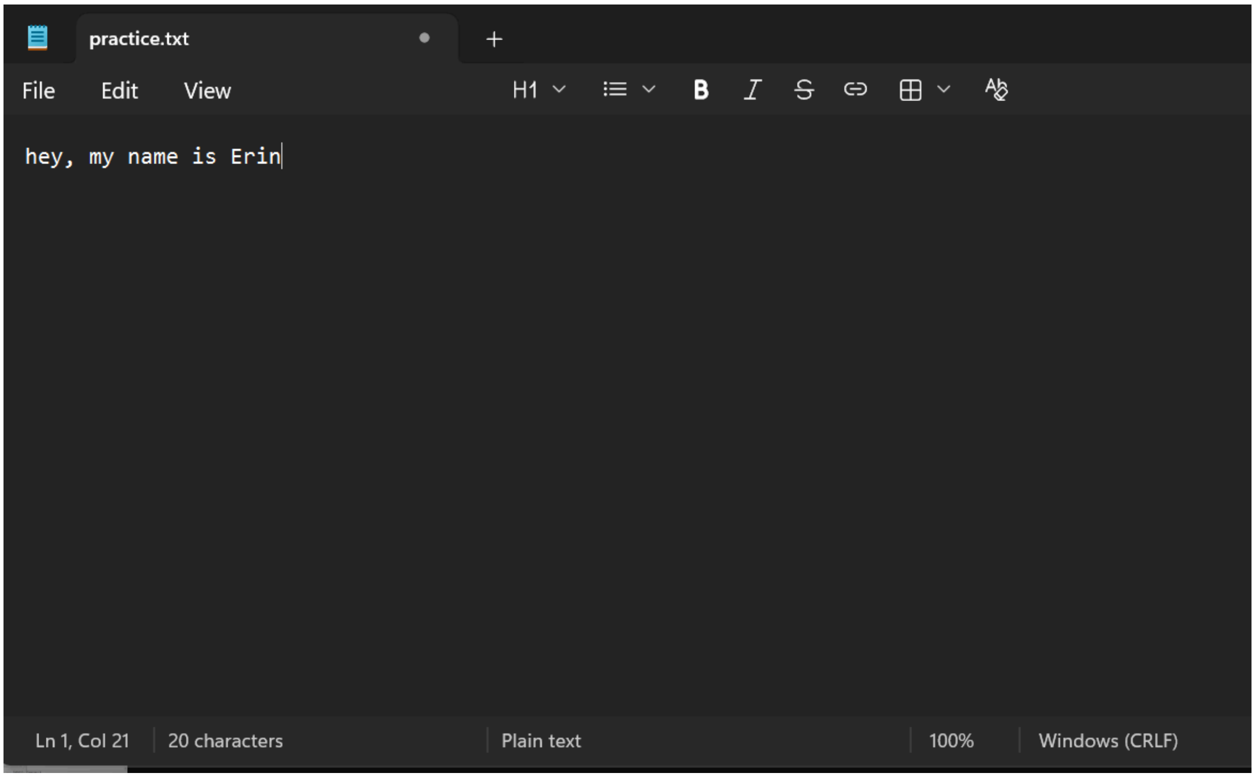
OBJECTIVES-
- Understand File System Behavior
- Identify Metadata vs. Content
- Differentiate File Statuses
- Perform Non-Destructive Recovery
- Verify Data Integrity

TOOLS REQUIRED-
- FTK Imager (Exterro)
- Windows Notepad
- Windows OS (File Explorer)
- Drive (D: Drive)

THEORY-
Digital forensics relies on the principle of Data Persistence, which states that data is rarely erased immediately upon deletion. In the NTFS file system, when a file is deleted, the Operating System does not wipe the physical bits from the disk; instead, it simply marks the file's entry in the Master File Table (MFT) as "available" for new data. This creates Unallocated Space—sectors on the drive that contain the original data but are no longer protected by the OS. Forensic tools like FTK Imager bypass the high-level Operating System view to read these low-level pointers and hex data, allowing investigators to identify, preview, and "Export" (copy) this data before it is overwritten by new system activity.

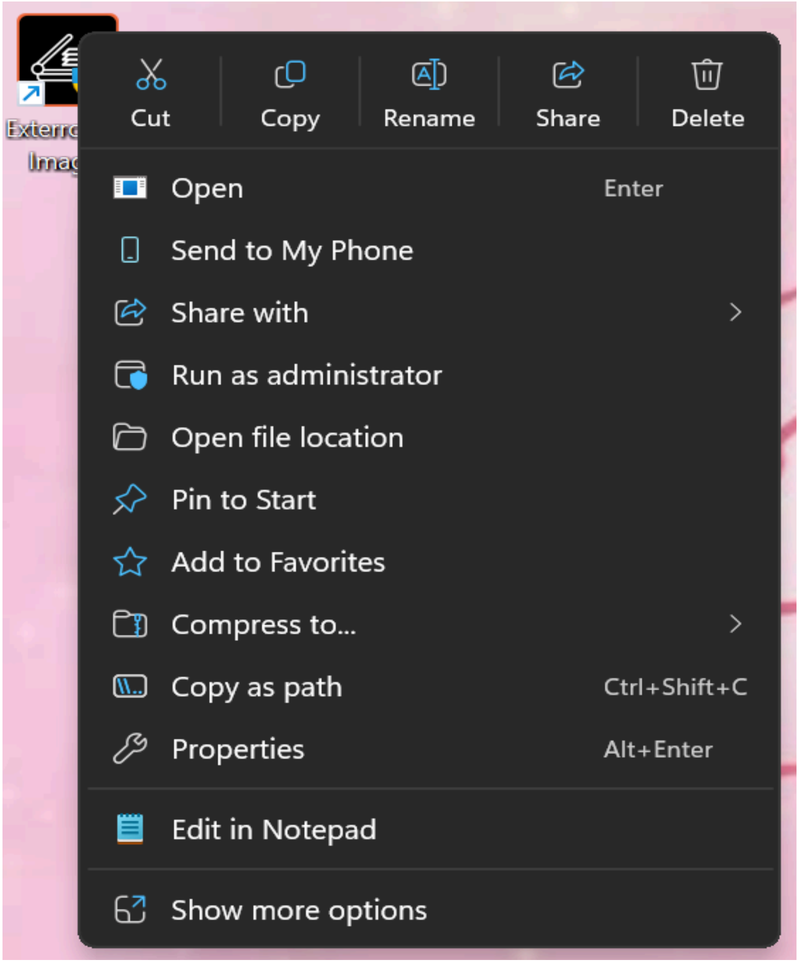
IMPLEMENTATION-

STEP 1: PREPARATION OF A FILE IN THE NOTEPAD AND DELETE IT AFTER SAVING



STEP 2:LAUNCHING THE TOOL

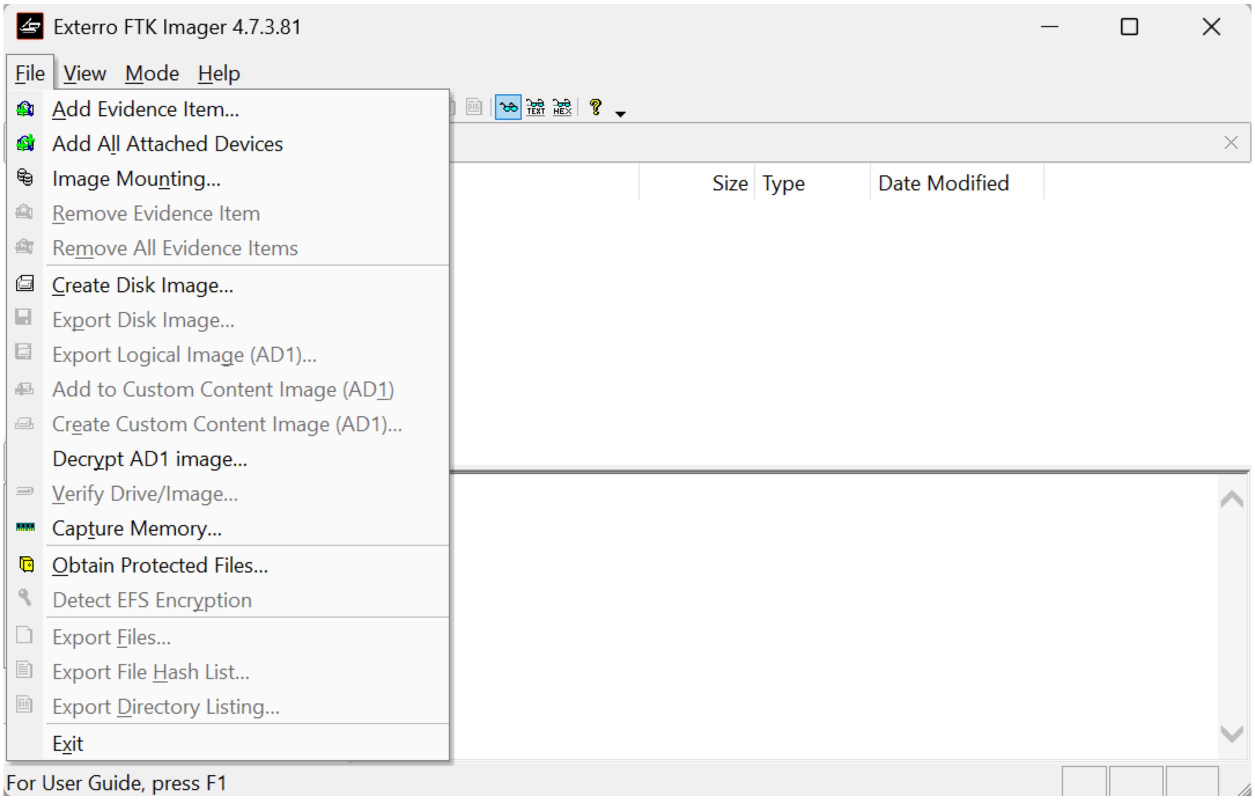
Right-click FTK Imager and select Run as Administrator. This is required to access the raw data of the drive.

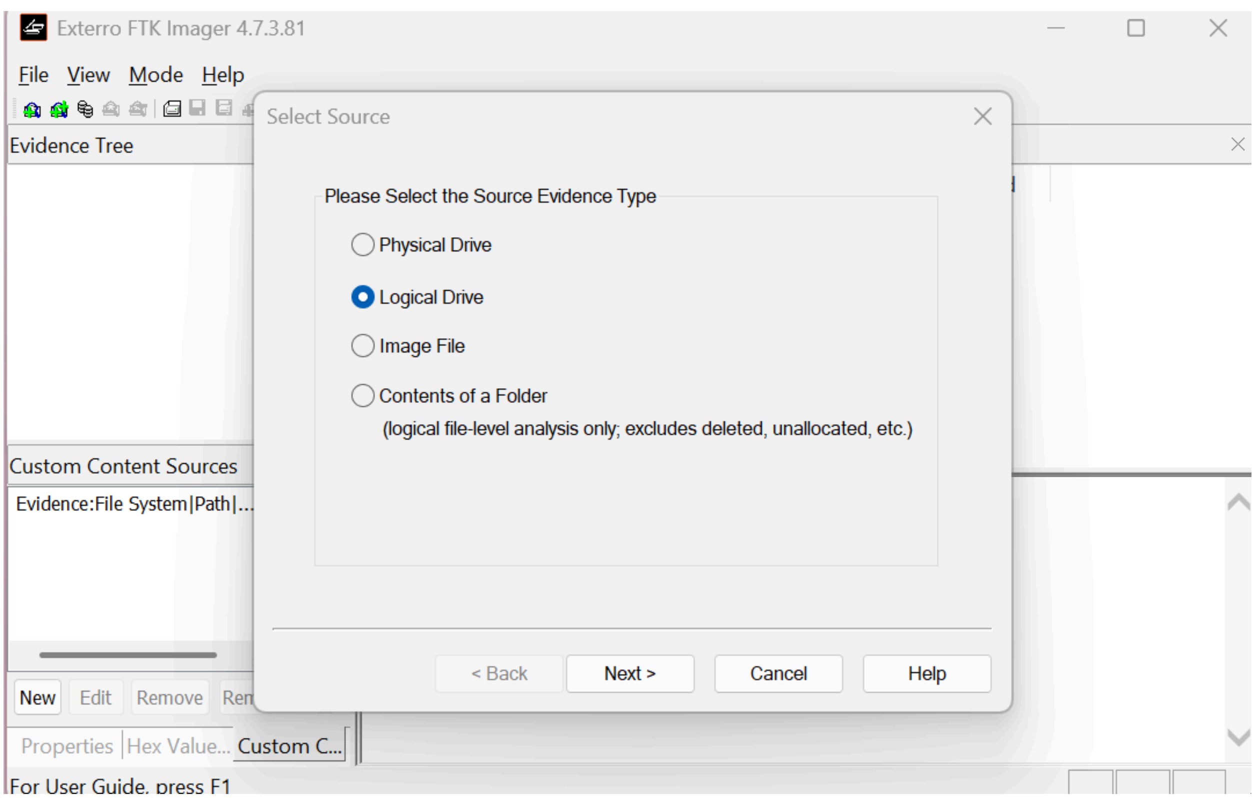


STEP 3-LOADING THE EVIDENCE

Go to File > Add Evidence Item.

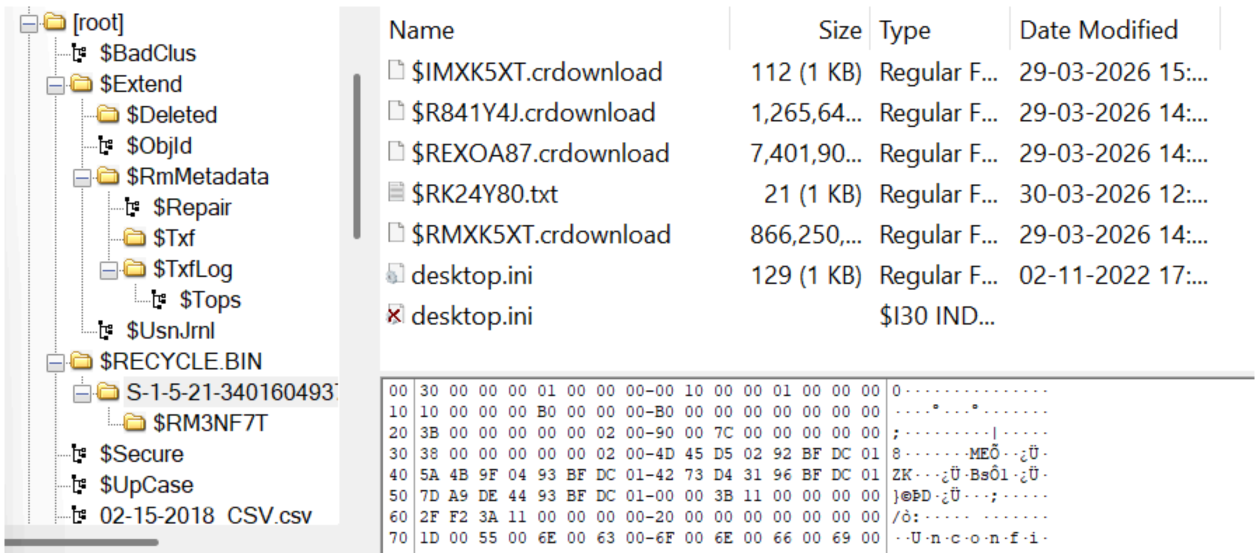
Select Logical Drive and choose the D: Drive from the dropdown menu.
Click Finish.


Exterro FTK Imager 4.7.3.81
File View Mode Help
Add Evidence Item...
Add All Attached Devices
Image Mounting...
Remove Evidence Item
Remove All Evidence Items
Create Disk Image...
Export Disk Image...
Export Logical Image (AD1)...
Add to Custom Content Image (AD1)
Create Custom Content Image (AD1)...
Decrypt AD1 image...
Verify Drive/Image...
Capture Memory...
Obtain Protected Files...
Detect EFS Encryption
Export Files...
Export File Hash List...
Export Directory Listing...
Exit
Size Type Date Modified
For User Guide, press F1


Exterro FTK Imager 4.7.3.81
File View Mode Help
Evidence Tree
Custom Content Sources
Evidence:File System|Path|...
New Edit Remove
Properties Hex Value... Custom C...
For User Guide, press F1
Select Source
Please Select the Source Evidence Type
Physical Drive
Logical Drive
Image File
Contents of a Folder
(logical file-level analysis only; excludes deleted, unallocated, etc.)
< Back Next > Cancel Help

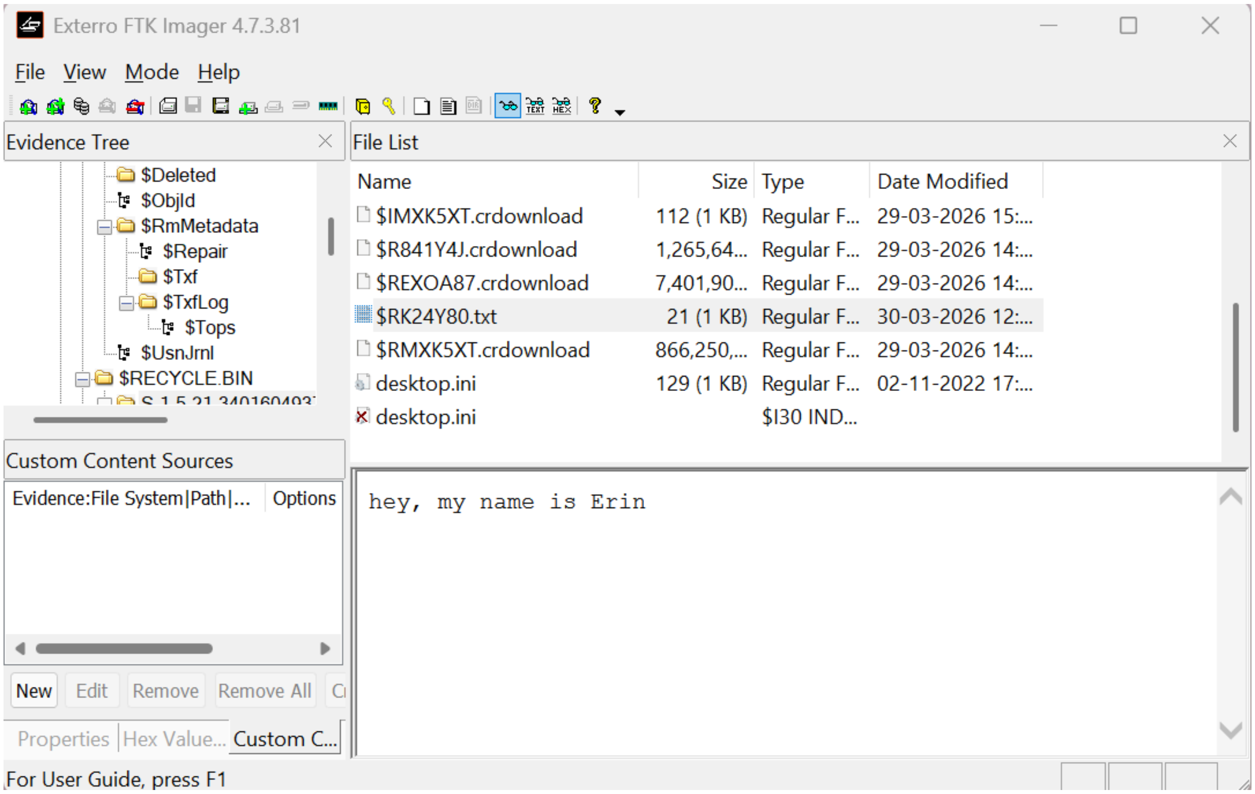
STEP 4-DATA DISCOVERY

The file with the red cross was deleted from the recycle bin. Here the .txt file was the deleted file.



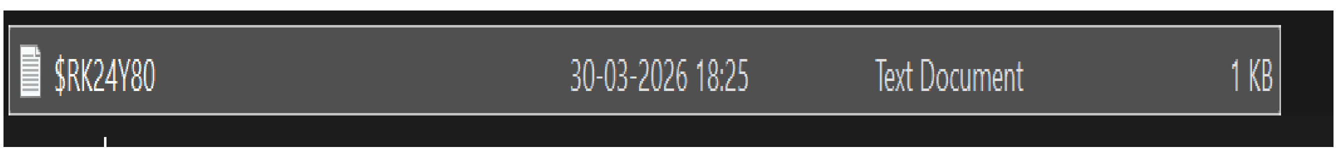
STEP 5-DATA VERIFICATION

Click on the suspected file and view its content in the Content Viewer (bottom pane). Switch to the Text tab or Hex tab to confirm the unique sentence matches your original file.



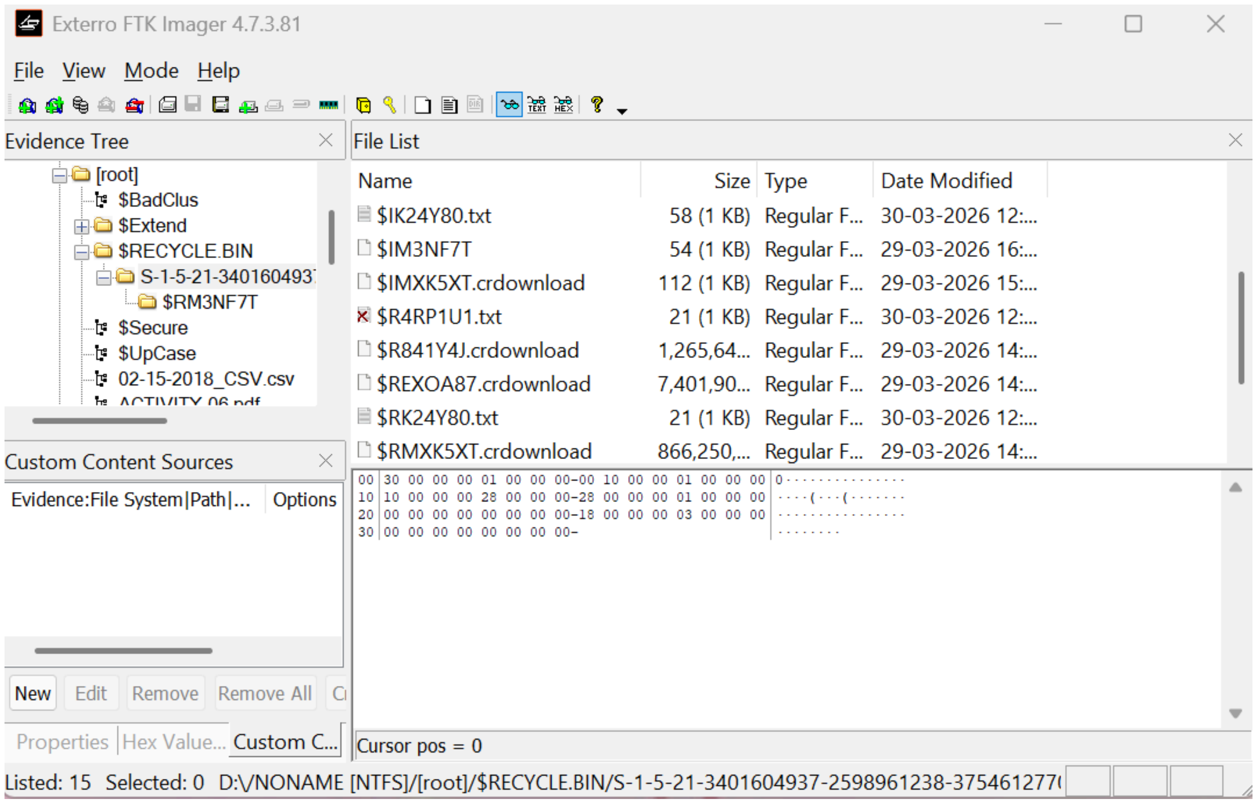
STEP 6- FORENSIC RECOVERY

Right-click the identified file and select Export Files.



The file recovered back into the drive after it was exported.

STEP 7-DELETING THE FILE FROM THE RECYCLE BIN



The .txt file is now marked with red cross as its been deleted from the recycle bin.

ALGORITHM-

- Initialize Environment
- Mount Evidence
- Scan for Deletion Markers
- Verify Data Integrity
- Execute Recovery (Export)
- Validation

OBSERVATION-

- File Renaming in $RECYCLE.BIN
- Presence of Deletion Markers
- Data Persistence in Hex View
- Successful Logical Recovery

RESULT-

The experiment concluded with the successful forensic recovery of a deleted notepad file from drive using FTK Imager.